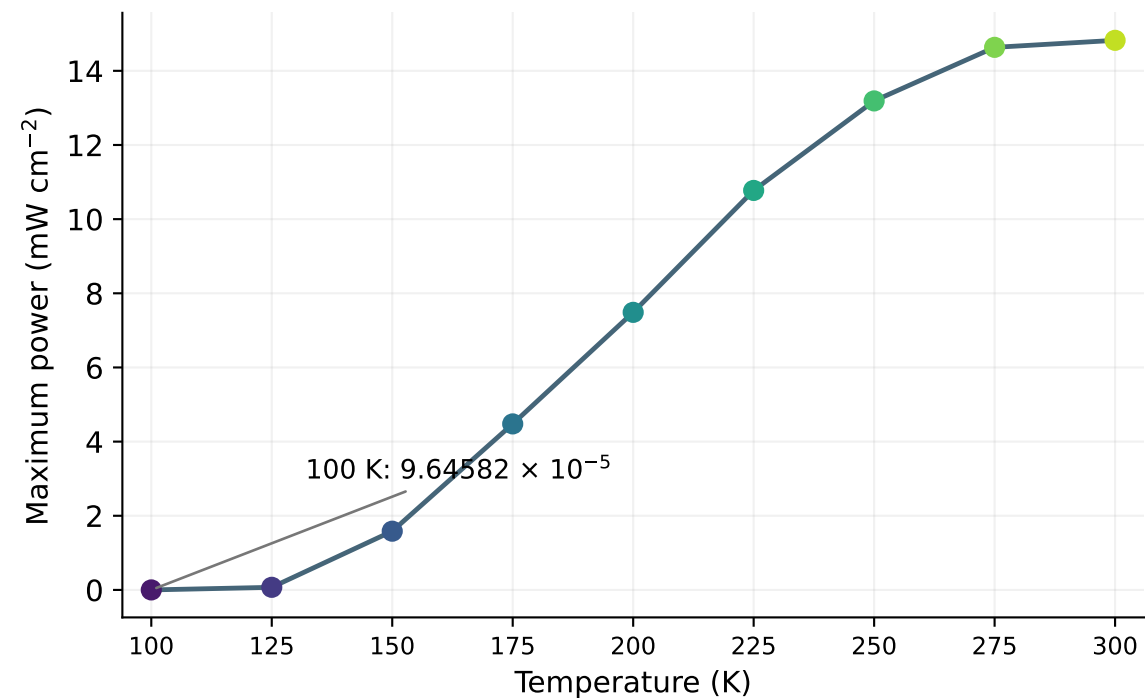
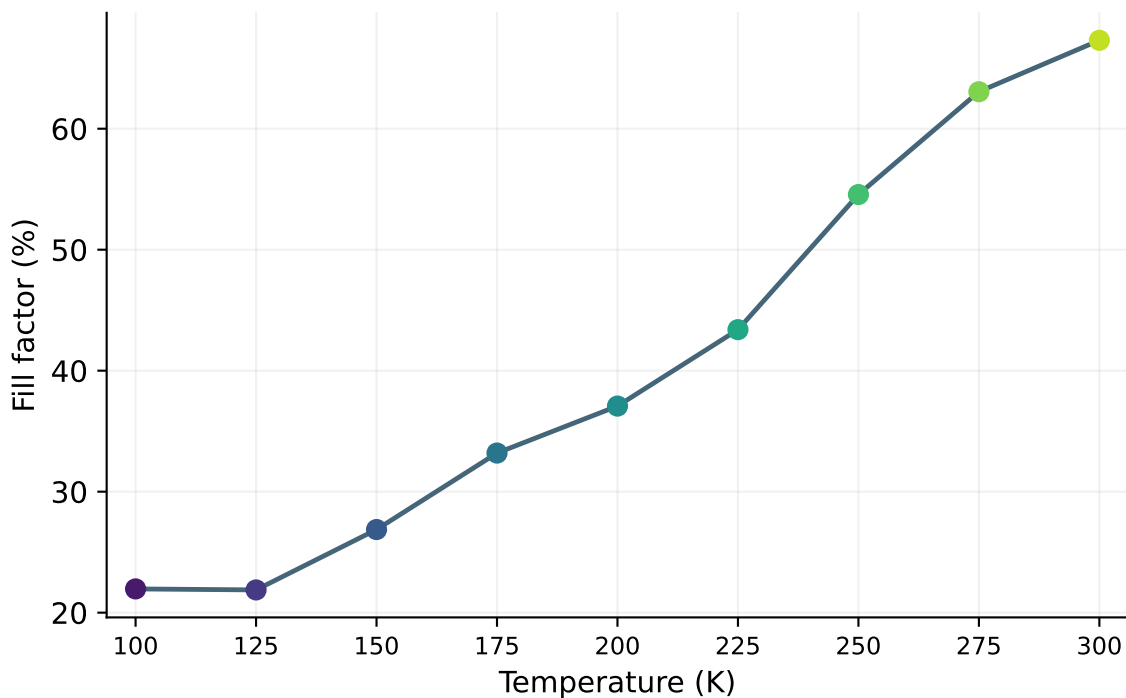
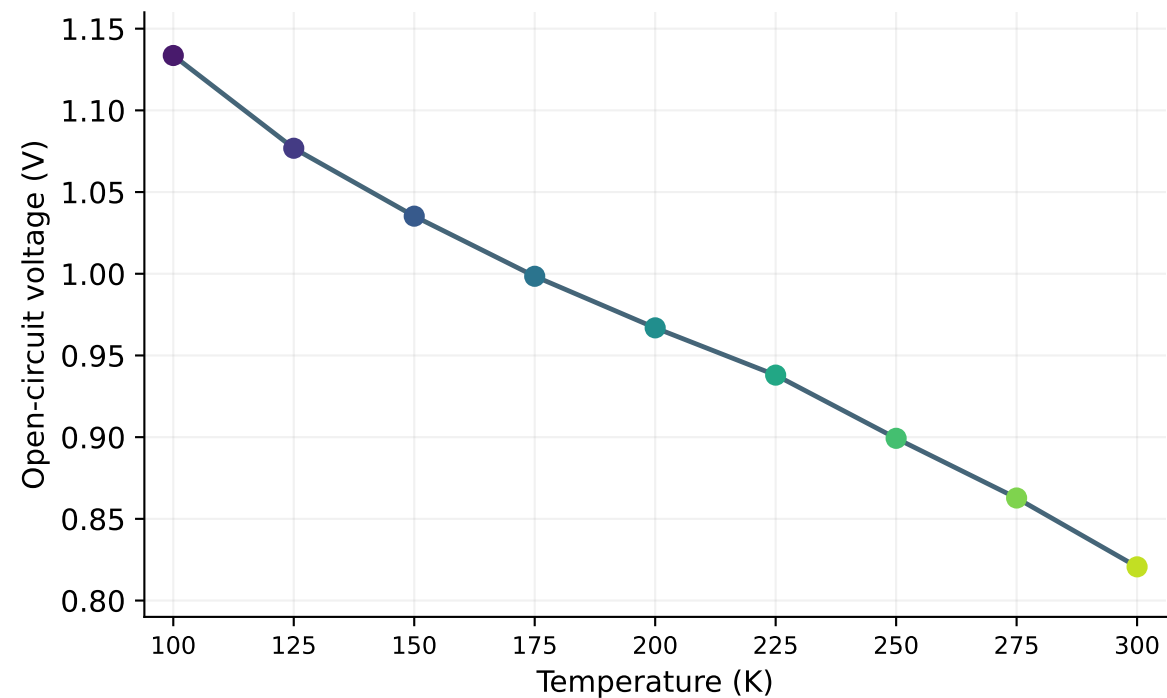
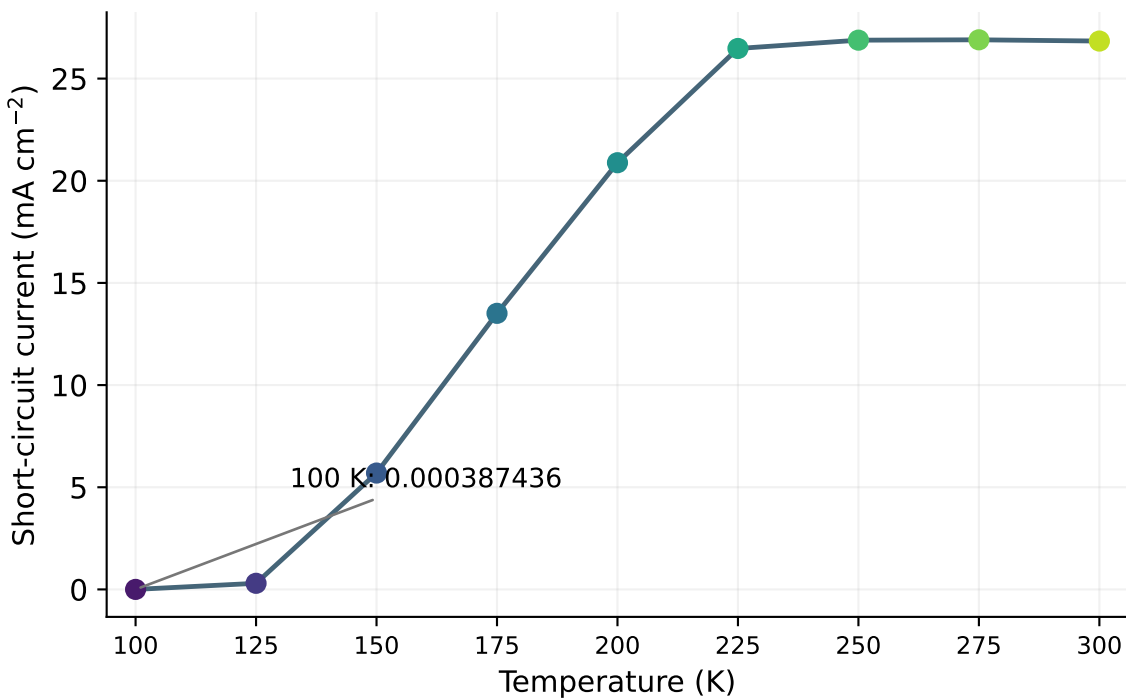


Photovoltaic metrics | no external shunt | all axes linear



N = 321 primary curves. Removed only $J_{sh} = V/R_{sh}$; series resistance remains zero.
300 K calibration parameters retained without refitting; low-temperature predictions remain conditional.